COMPUTATIONAL PHILOLOGY · CORPORA · NEGATIVE RESULTS

Reading Steppe Names in Chinese Transcription

Three corpora, two medieval lexicons, and a measured answer to a fifty-year-old question about how much chance a sound law may tolerate.

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Preprint. Data and code: github.com/ulug2004/chinese-transcription-channel

ABSTRACT

Chinese histories record hundreds of Inner Asian names (Xiongnu rulers, Turkic titles, tribal designations) in characters chosen for their sound. Two thousand years of change in Chinese pronunciation have left the modern Mandarin readings of those characters unrelated to what the scribes heard. This paper models the transcription process itself as a noisy channel, trains it where both sides are known, and measures what can and cannot be recovered.

We build and release **three transcription corpora**, two of them extracted from sources not previously available as data: 9,329 Chinese–Mongolian pairs from the *Secret History of the Mongols*, 594 Chinese–Uyghur pairs from Ligeti's edition of a Ming glossary, and 1,017 verified Chinese–Sanskrit pairs separated out of a Buddhist dictionary that mixes phonetic spellings with translations; and **two medieval Turkic lexicons** in machine-readable form, for which we found no downloadable dataset (§3). On held-out data the channel reconstructs 31% of Turkic words exactly and 84% within two segments (§5).

Three results follow. **One positive reading** survives every test we can apply: 若鞮, the element closing the formal names of six later chanyu, reads as *İnakt* "the faithful ones" (§9). **One century-old disagreement is relocated**: the dispute over 冒頓 turns on which value is taken for a single character, which carries two Later Han readings and a third from a commentary. We price each branch, and no published reading of this name states the choice it is making (§9.2). And **a set of measured negative results**, one of them demonstrated on an item chosen by the Yeniseian hypothesis rather than by us. Language attribution from transcription evidence is not supportable on available reference data: at 閼氏, the chanyu's consort title, the same spelling writes a well formed Yeniseian *elte* "wife" and a Turkic *elti* "lady" by exactly the same steps, so it licenses both and selects neither. Nor is full reconstruction supportable at the density of evidence that survives: 1.4 observations per character, against the 60 that made the same method work on Mongolian (§6, §9.1).

Section 7 answers the question the field has carried unresolved since 1973. Doerfer argued that chance resemblance is pervasive in long-range comparison; his reviewer sharpened it into a threshold problem: *is it a sound law if 49% of cases show one correspondence and 51% another?* We apply Doerfer's three classes with confidence intervals, and find that the threshold cannot be a constant: it is the baseline of the component being measured, and it moves with how many outcomes that component has.

Keywords: historical phonology · noisy-channel models · corpus construction · Xiongnu · Old Turkic · null models · negative results

1 The problem

When a Chinese scribe of the Han dynasty wrote down a foreign name, he chose the characters for their sound. 冒頓, the founder of the Xiongnu state, is read *Mòdùn* in modern Mandarin, but Mandarin is not the language the characters were chosen in. In the Later Han period those two characters were pronounced approximately as `mək tuənᶜ`.[22] Every argument about what such a name meant, and about who the people bearing it were, depends on getting from the modern reading back to the period one.

That is a recoverable step: reconstructions of Later Han Chinese are published. The question is given a period reading, can the original foreign word be recovered. The transcription is lossy: Chinese had no `r-`, a restricted set of syllable shapes, and no way to write many foreign sounds. So, the mapping from source word to spelling is many-to-one and its inverse is underdetermined. This is a noisy-channel problem, and it can be treated as one.

The interest of the Xiongnu case is that the answer is unknown and contested: proposals put the ruling house's language in the Turkic, Mongolic, Yeniseian and Iranian families, each resting on a handful of transcribed names; the most recent argues for Yeniseian, specifically an early form of Arin,[3] and Doerfer argued half a century ago that none of the proposals could be sustained on the evidence then available.[6] The interest of this *method* is that its reliability can be measured in cases where the answer is known, which is what most of this paper does.

2 A question the field has already asked

Nothing in the approach below is methodologically new. Gerhard Doerfer made the argument in 1973, in *Lautgesetz und Zufall* ("Sound Law and Chance"), a study of what he called *omnicomparatism*: relating languages across great distances on the strength of resemblances.[5] Chance produces such resemblances in quantity, was Doerfer's case, and comparativists have a repertoire of devices (he called them tricks) for making weak matches look strong. He included deliberately absurd comparisons of his own to show how easily the devices work.

A review in *Anthropos* the following year pressed the point into a question that can be answered with numbers.[15] Sound laws, the reviewer observed, are not deterministic but probabilistic, and the literature never says how probabilistic they are permitted to be. *Is it a sound law if 49% of cases show the same correspondence and 51% show something else?* Where does regularity stop? He proposed three classes: deterministic laws, holding without exception; strong probabilistic laws, where one correspondence occurs in more than half of cases; and weak probabilistic laws, where one correspondence is merely the most common, and could apply none of them, for want of counts.

He also named the mathematics that was already available: the matching problem of Montmort (1708), which asks how often two shuffled sequences agree by accident. The permutation nulls used throughout this paper are that problem solved by simulation rather than by formula. What is new here is not the method but the measurement: we compute the numbers on this material instead of asserting them, which is what Doerfer was asking for.

3 Resources

A learned model needs examples, and for this problem almost none existed as data. We built five datasets as the needed resources: three of transcription pairs, and two lexicons of Turkic itself. We release or make reproducible all of them (see §12).

3.1 Corpus 1: *Secret History of the Mongols*

The *Secret History* is a thirteenth-century Mongolian chronicle preserved only in Chinese characters used phonetically, with an interlinear gloss. It is the largest known parallel text of its kind, and it had not been extracted as aligned pairs. Parsing the digital edition yielded **9,329 Chinese–Mongolian transcription pairs**.

One detail decides whether the extraction worked. In the source encoding, a Chinese character followed by U+180B (a Mongolian free variation selector) marks an *annotation*, not part of the transcription. Treating those as data corrupted roughly a fifth of the corpus; the rule is one line, and finding it took considerably longer. Length agreement between the Chinese and Mongolian sides is 93.4%, which is the internal check that the alignment is real.

3.2 Corpus 2: Ligeti's Sino-Uyghur vocabulary

Ligeti[10,17] published a Ming-dynasty Chinese–Uyghur glossary in two instalments in *Acta Orientalia*.[17,18] The Turkic forms are given in Latin transcription and the Chinese in EFEO romanisation, which makes the text parseable. Extraction yielded **594 Turkic–Chinese pairs**.

We had to manage four properties of the typography: the apostrophe is U+2019 rather than ASCII, Turkic forms may contain spaces, footnote digits intervene before the gloss, and the Turkic form is not always at the start of a line. Our first pass missed them and returned

297 pairs, half of them wrong. A further trap was structural rather than typographic: journal running heads carry the author's name on verso pages, so a filter written for the article title alone silently discarded half the article.

3.3 Corpus 3: verified Buddhist Sanskrit transcriptions

Buddhist translators into Chinese used two devices: **音譯**, spelling a Sanskrit word by sound, and **意譯**, translating its meaning. Only the first is transcription. A 27,110-entry terminology dictionary[1] mixes them without marking which is which, and the two are not separable by inspection.

We had two approaches fail before one worked. Character-set membership admits **福慧** for *puṇya*, which is a calque. Exclusivity, treating a character as phonetic only if it never appears in a translation, collapsed the corpus to 67 pairs. What worked was phonetic verification of scoring each candidate against the Later Han reading of its characters and keeping only those whose sound matches. The result was **1,017 verified pairs**. We kept the 7,480 rejected entries separately so anyone can audit the decision. The induced **音譯** inventory matches the standard akṣara table, which is an external check providing that the filter found something real.

3.4 Lexicons 1 and 2: Kāşgarî and the *Codex Cumanicus*

The tests in §6 need dictionaries of Turkic itself, from near the period, against which a reading can be looked up. We found no downloadable dataset, so we extracted the following dictionaries.

Kāşgarlı Mahmud's *Dīwān Lughāt al-Turk* (1072–77) is the earliest dictionary of any Turkic language.[12] The Türk Dil Kurumu index volume has a machine-readable text layer. Parsing it yielded **6,820 headwords** with Kāşgarî's Turkish glosses and his volume-and-page references.

The *Codex Cumanicus* (c. 1300) records Kipchak Turkic, compiled by Italian merchants and German missionaries, outsiders writing down a language they did not speak, which is the position the Chinese scribes were in. Géza Kuun's 1880 edition prints a Cumano-Latin vocabulary. Parsing it yielded **2,110 headwords** with Latin glosses.[14] We deliberately skipped the Persian and German vocabularies that follow in the same volume. The Persian one would inject Iranian vocabulary into a Turkic lexicon and inflate every match rate computed from it.

4 Method

The channel we built is a probabilistic model of the Chinese scribe: for a source sound, what Chinese spelling would he choose? Character-to-sound alignments are learned by expectation–maximization over the training pairs, without hand-alignment. Emission tables are *reading-anchored* with backoff: full reading, then onset plus rime, then onset, then a global distribution, so a character never seen in training still receives a distribution rather than nothing.

Decoding runs the channel backwards: given the characters, score candidate source words. The binding variable throughout is not corpus size but **observations per unit**: how many times the model has seen each thing it must estimate. That figure, not the number of pairs, is what predicts whether reconstruction works.

We paired every comparison in this paper with a null. For lexicon searches we built pseudo-names by resampling Later Han syllables from the same names being tested, so they carry the right shape and the right sound inventory and no meaning at all; the identical search is run on them. For language comparisons we subsample the corpora to equal size before scoring, because otherwise the thinner model wins by being vague. Negative controls, meaning the method run on cases whose answer is already known, are what exposed the failures reported in §6.

5 What channel recovers

Every figure below is on held-out data. "Exact" means the model reconstructed the whole word correctly; "within 2" means it got no more than two segments wrong.

TRAINED ON → TESTED ON	EXACT	WITHIN 1	WITHIN 2
Mongolian → Mongolian (unseen chapters)	36.9%	66.3%	87.5%
Turkic → Turkic	30.6%	68.2%	83.5%
Sanskrit → Sanskrit	25.0%	57.9%	77.0%
Sanskrit → *Turkic* (wrong training language)	3.1%	6.9%	27.5%

TABLE 1 Each row is the result of one experiment. Row 1 trains on some chapters of the *Secret History* and tests on chapters held back entirely, a document-level split, so no word from a test chapter influenced training. Rows 2 and 3 do the same within the

Turkic and Sanskrit corpora, but those are glossaries without document structure, so the held-out portion is a random sample of pairs. Row 4 is the diagnostic: the channel is trained entirely on Sanskrit and asked to read Turkic. A model trained on Buddhist transcriptions barely functions on Turkic ones, 3.1% against 30.6%, which is the clearest evidence that the channel learns something specific to the language it was trained on rather than a generic Chinese-to-anything mapping.

CORPUS	PAIRS	OBSERVATIONS PER CHARACTER
Mongolian	9,329	~60
Sanskrit	1,017	~12
Turkic	594	~7
Xiongnu names	39 items	1.4

TABLE 2 The Xiongnu material is 112 character-tokens over 81 distinct characters. This is the number that governs everything in §6: the method does not fail on Xiongnu because Xiongnu is mysterious, but because 1.4 observations per character is not enough to estimate anything. The comparison is with the ~60 of the Mongolian corpus, which is the density at which the same method reconstructs 36.9% of words exactly (Table 1).

6 What the evidence cannot support

Two claims commonly made from this kind of material do not hold up and establishing that carefully is important.

6.1 Determining which language a name belongs to

The obvious application is settling whether the Xiongnu ruling house spoke Turkic, Yeniseian, Mongolic, or something else. We built the comparison properly: score how well each candidate language explains the transcription, equalise corpus sizes, compare by Bayes factor.

It failed its own controls. Given Ming Uyghur transcriptions whose language is known as Turkic with certainty, the method identified Turkic correctly in 69% of trials against a 50% chance baseline, and in 2 of 24 experimental cells it fell *below* chance. A diagnostic showed why: the phonotactic priors available for these families, taken from ASJP,[29] are close enough to each other that the comparison is measuring the reference data rather than the names. A procedure that cannot reliably identify a language it has been given cannot be trusted to identify one nobody has.

The strongest advocate of the Yeniseian hypothesis reaches the same conclusion on methodological grounds rather than measured ones. Vovin, arguing for a Yeniseian Xiongnu, sets the titles and personal names aside as evidence altogether: most of the glosses the Chinese sources preserve are of that kind, he calls them "practically useless" for determining genetic affiliation, and he declines to evaluate the ones Pulleyblank had used, on the ground that such evidence is not admissible.[27] The record this paper works from is almost entirely titles and personal names. Our result agrees with his judgement and replaces it with a number.

6.2 Reconstructing the Xiongnu names in full

At 1.4 observations per character the decoder returns candidates, but they are not constrained by evidence. An intermediate version of this work gated its output on in-domain accuracy and would have printed confident readings (**撑犁** as *ara*, *kara*, *tara*) where true out-of-domain reliability was 3.1%. Re-gating on out-of-domain accuracy removed them. This was the closest the project came to publishing a wrong answer, and we record the correction here because the failure mode is general: a decoder always returns its best candidate, and best is not the same as supported.

6.3 Matching readings to dictionary words

There is an obvious next step, and it is the one every reader wants to take. Once **稽粥** is written out as *kei çuk*, it is hard not to hear Turkish *küçük* "small, younger". The method is simple: look the reading up in a dictionary and see what it matches.

The trouble is that it always works. Modern Turkish has twenty thousand words in common use, and a target with three consonants has hundreds of near neighbors. The question is never "is there a match" but "is there more of a match than a string that cannot mean anything would get?" We built seven lexicons and gave each its own null.

The modern side of the comparison uses open spelling dictionaries rather than a published dictionary's headword list because a spelling dictionary can be counted. Turkish is the *hunspell-tr* word list, whose 292,000 forms are cut to 21,721 by keeping only words with a wordfreq Zipf score of at least 2.5, that is, words a speaker actually uses.[25,23] Kazakh is the *myspell-kk* list, 53,182 forms, romanised on load by a fixed Cyrillic-to-Latin table.[13] The frequency filter matters: without it the Turkish list is dominated by inflected and rare forms, and every target finds a match. The historical side is described in §3.4.

LEXICON	ENTRIES	PSEUDO-NAMES FINDING A MATCH	MEDIAN SCORE FOR A PSEUDO-NAME	NAMES BEATING CHANCE, $p \le 0.05$
Turkish, common use	21,721	231 / 250	0.156	2 of 36
Kazakh	53,182	239 / 250	0.183	0 of 36
Union of Turkish and Kazakh	73,406	236 / 250	0.117	2 of 36
Turkish, loan-shaped words removed	14,876	233 / 250	0.200	2 of 36
Intersection (in both languages)	1,497	219 / 250	0.329	3 of 35
Dīwān Lughāt al-Turk, 1072–77	6,337	276 / 300	0.200	3 of 36
Codex Cumanicus, c. 1300	2,067	275 / 300	0.242	3 of 36

TABLE 3 A lower score is a closer match, so a *lower* median in column four means the lexicon is noisier: meaningless strings are finding better matches in it. Read that column as the price of admission. Two results are worth separating out. **Pooling two dictionaries makes matters worse**: the union finds more matches for the real names and more for the meaningless ones, and the noise floor falls from 0.156 to 0.117. A larger haystack does not help you find a needle. What helps is the opposite operation of keeping only words present in *both* languages. This lifts the floor to 0.329, because a word must survive two independent borrowing histories to qualify. Unfortunately, the intersection is over identical spellings, and cognates diverge, so Turkish *tanrı* and Kazakh *täñir* both fall out while *domino* and *kauçuk*, borrowed into both recently, survive. This enriches for shared modern loan words. Across all seven lexicons, between none and three names in thirty-six could meet $p \le 0.05$ significance, which at a one-in-twenty threshold is what thirty-six coin flips give.

The older lexicons do better than modern ones on two cases with known answers (*tümen*, *tengri*).

NAME	READING	LEXICON	RANK	CANDIDATE	GLOSS IN THAT DICTIONARY
撐犂 chengli	ḍaŋ lei†	Modern Turkish	–	not returned	*dengeli* "balanced" came first
		Dīwān Lughāt al-Turk	4th	tengri	*gök, sema* ("sky, heaven")
		Codex Cumanicus	2nd	tengri	*deus* ("God")
頭曼 Touman	do mɑnᶜ†	Modern Turkish	–	duman	"smoke"; *tümen* not in the top three
		Dīwān Lughāt al-Turk	3rd	tümen	*pek çok* ("very many")
		Codex Cumanicus	2nd	touman	*nebula* ("mist"); *tümen* absent from this lexicon

TABLE 4 The *Hanshu* glosses 撐犂 as 天 "heaven". Kāşgarî in Baghdad and the compilers of the Codex in the Crimea, working a thousand years later with no knowledge of the Chinese record, gloss *tengri* as "sky, heaven" and as *deus*. The attestations are independent and the search connects them, which fails to do against a modern dictionary, and does better the older and smaller the lexicon becomes. The second name repays attention because it cuts against the accepted etymology: *tümen* "ten thousand" is absent from the Cuman vocabulary entirely, and in every lexicon holding both, the search prefers a different lexeme: Turkish *duman*, Kāşgarî's *tuman*, Cuman *touman*, all "smoke, fog, mist", one word across three lexicons and a thousand year separation. The difference is one vowel, and §7 puts vowel recovery at 50%. This is recorded because the alternative has never been raised.

A caution about the strongest datum. The match between ḍaŋ lei and *tengri* is the most quoted evidence for a Turkic Xiongnu. The inference only works if *tengri* is Turkic to begin with. The word is not confined to Turkic: Mongolic has *tenger* "sky", Turkic *tengri* survives as modern Turkish *Tanrı*, and Georg has argued that the Turko-Mongolic word is built on Proto-Yeniseian *təŋgVr- "high" with a Turkic suffix, making it a borrowing rather than an inheritance.[8,28] Georg's derivation is one argument among several and need not be accepted; what holds either way is that a word attested across Turkic, Mongolic and, on Georg's account, Yeniseian cannot discriminate between them. 撐犂 shows that the Xiongnu used, for "heaven", a word also found in Turkic and Mongolic. It does not show which of those languages they spoke.

7 Answering the 1974 question

The Xiongnu names survive in Han-period sources; the Turkic training corpus we used is Ming era, which is twelve hundred years later. We can measure directly whether anything carries across that gap. We took characters whose readings are known in both layers, predict the earlier reading from the later one, and score by component.

PART OF THE SYLLABLE	RECOVERED CORRECTLY MING → LATER HAN	95% CONFIDENCE INTERVAL	CHANCE BASELINE: BEST BLIND GUESS	LIFT: POINTS GAINED OVER THAT GUESS	CLASS OF SOUND LAW[15]
Nasal ending (-m, -n, -ng)	99.2%	95.4 – 99.9	35.4%	+63.8	deterministic
Initial consonant	79.3%	72.4 – 84.8	9.1%	+70.1	strong probabilistic
Vowel	50.0%	42.4 – 57.6	15.2%	+34.8	weak probabilistic
Stop ending (-p, -t, -k)	0.0%	0.0 – 7.9	35.4%	−35.4	no law

TABLE 5 n = 164 characters (nasal ending 118, stop ending 45). "Recovered correctly" is leave-one-out majority prediction of the earlier reading from the later; intervals are Wilson score intervals; the chance baseline is the unconditional majority class for that part, the score obtained by always guessing the commonest value. **Lift is column two minus column four**: how many percentage points knowing the later reading provides. The aggregate coda figure shouldn't be used. Blending nasals and stops gives a flattering 87%, which means nothing, because the sample is 72% nasal-ending characters.

This is the reviewer's question, answered. Three of the four components classify cleanly. Nasal endings, at 117 of 118, are deterministic for practical purposes. Initial consonants, at 79.3% with the interval nowhere near one half, are a strong probabilistic law. Stop endings are not a law in any sense: not one of 45 survived, and the interval tops out at 7.9%.

The fourth exposes a flaw in the rule itself. Vowels land at **50.0%** (82 of 164, on the boundary to the decimal), and the confidence interval, 42.4 to 57.6, straddles it. By the reviewer's criterion the vowel cannot be classified, and no larger sample would help, because the estimate is certain; it is genuinely at one half.

But one half is the right threshold only when there are two outcomes. A vowel is a choice among many vowels, and the honest comparison is against what you would get knowing nothing, the majority class, here 15.2%. Against that, vowels carry **+34.8 points** of real information: While it is a weak probabilistic law, it is not a coin flip. Stop endings, by the same measure, fall *below* their baseline. Knowing the Ming reading makes you worse off than guessing, because Old Mandarin merged -p, -t and -k into nothing and the merger is not recoverable.[24]

What this changes about the question. The reviewer asked for one number: the percentage above which a correspondence counts as a sound law. The four components show that no single number can serve, because the same percentage means different things in different places. A vowel correspondence at 50% and a coin landing heads at 50% are not the same result. The coin has two outcomes, so 50% is exactly what ignorance gives you and the correspondence carries no information at all. The vowel has many outcomes, and ignorance gives you 15.2%, so the same 50% is more than three times the rate you would get by knowing nothing, and it carries a great deal. The threshold that separates a law from chance is therefore not a constant to be agreed on once and applied everywhere. It is the baseline of the component being measured, it is set by how many outcomes that component has, and it has to be computed for that component before any percentage can be read at all. That is why the question could stay open for fifty years without either side being wrong: it was posed as a question about percentages, and it is a question about baselines.

8 Five measurements of scribal practice

8.1 Is a Chinese `m-` ever a foreign `b-`?

Reading 冒頓 as *Bayatur*, the most repeated etymology in the field, requires the `m-` to stand in for a `b-`. Proto-Turkic has no initial `*m-`: the `m-` words in Old Turkic are loans or come from `*b-` assimilating to a nasal immediately after the vowel: **bän* → *men*, **buŋ* → *muŋ*.[4,7] So on a Turkic reading the whole etymology rests on that substitution being something a Han scribe would do. Each corpus is a set of transcriptions whose source word is known, which makes the question a counting exercise.

CORPUS	SOURCE WORDS BEGINNING `B-`	WRITTEN WITH A CHINESE `M-` CHARACTER
Mongolian (*Secret History*)	738	0
Turkic (Ligeti glossaries)	78	0

CORPUS	SOURCE WORDS BEGINNING `b-`	WRITTEN WITH A CHINESE `m-` CHARACTER
Sanskrit (Later Han, verified)	22	3 (13.6%)
Total	**838**	**3** (0.4%)

TABLE 6 The reverse direction is clean: Chinese `m-` characters render a source `m-` in 303 of 305 Mongolian cases, 90 of 97 Sanskrit and 14 of 17 Turkic. The forward direction, which is the one that matters here, must be read by layer and not pooled. The total of 3 in 838 is 0.4%, but all three instances fall in the *Later Han* corpus, where the rate is **3 of 22, or 13.6%**, with a Wilson interval of 4.7 to 33.3. The two later corpora contributed 0 of 816 between them. Pooling across eight centuries of Chinese produces a figure that describes no period at all, which is the point §7 makes in general terms.

Read by layer, the substitution is not exceptional. A rate of 13.6% is the same order as `ts-` for a source affricate, which §6.3 treats as a minority but attested spelling at 15%. The scribe did have the alternative: within the Later Han corpus a source `b-` took a `b-` character 26 times in 32. But having an alternative and always using it are different things, and once in seven he did not. **So, an `m-` character is not, by itself, an argument against a Turkic reading of 冒頓.**

The onomastic record points the same way, in a later period. Tang scribes wrote the Türk ruler known in the inscriptions as Qapaghan Qaghan with the personal name **默啜**, read as *Bögü-Çor*; and the Uyghur ruler **牟羽** is the same word *bögü* "wise" behind a different `m-` character. Two different Chinese `m-` characters for one Turkic `b-` word is a pattern, not an accident. A third Türk name often cited alongside these two, **木杆** for Muqan Qaghan, is a control rather than a third instance: the Sogdian Bugut inscription writes that name *mwγʾn*, also with `m-`, in a script that has a letter for *b*, so the name's own initial is disputed and cannot be assumed to be `b-`. These are Tang rather than Han, and Late Middle Chinese treated its nasal initials differently, so they corroborate that the practice existed rather than fixing its Han-period rate; but they remove any sense that the substitution would be unheard of in a ruler's name.

The objection moves to the other end of the word. *Bayatur* ends in `-r`, and **頓** `tuənᶜ` carries a written `-n`. Later Han had no `-r` coda, so a scribe facing a source syllable such as *tur* had to choose something, and the Sanskrit corpus records what he chose. Of 37 source syllables ending in `-r`, he left the syllable **open** 21 times (57%), used a `-t` coda **14** times (38%): **達** for *-har* in *bodhidharma*, **羯** for *kar-* in *ākarṣaṇī*. A `-p` coda occurs once, and a `-n` coda **once**, 3%. Reading **頓** as *tur* therefore asks for the rarest of the four options, where the expected spelling was an open character or a `-t`.[22] The syllable count is no obstacle to either reading. *Bayatur* has three syllables against two characters, but 93 of the 1,017 verified pairs use fewer characters than syllables, and in 47 of those a character with a written stop coda does the compressing. **竭** covers *-gata* in **多阿竭** for *tathāgata*, **塞** covers *-saka* in **優婆塞** for *upāsaka*. Compression is a real device, available to any reading of **冒頓**, so the final `-r` stands alone as the difficulty.

What survives on the Turkic side is unaffected by any of this. Proto-Turkic has no initial `*m-`, and the one route to one is `*b-` assimilating to a nasal immediately after the vowel; **冒** `mək` has no nasal in that position, its only nasal being the `-n` of **頓**. So, on a Turkic reading the `m-` must be the scribe's choice rather than the source language's sound, and the count above (13.6%) says he was entitled to make it. Small numbers should be kept in view: the result rests on three transcriptions, and had those three gone the other way the rate would be zero. The interval, 4.7% to 33.3%, is correspondingly wide.

8.2 How was a source coda spelled?

A related question, and the one §9 turns on. When a source syllable ended in a consonant, did the scribe use a character ending in that consonant, or an open character with a vowel after it? The Sanskrit corpus answers it for the right period: for the 770 pairs aligning one character to one syllable, each Sanskrit syllable is matched to its character and the coda of each recorded.

SOURCE SYLLABLE ENDS	N	CLOSED WITH THE SAME CODA	WRITTEN WITH AN OPEN CHARACTER
-n	93	**83%**	13%
-ŋ	22	**73%**	9%
-t	53	53%	28%
-k	21	48%	38%
-p	10	20%	20%
All stops	**84**	**48%**	**30%**

TABLE 7 Roughly a third of stop-final source syllables were written with an open character, against 13% and 9% for nasals. An open spelling after a stop is ordinary, not exceptional. Note the shape of the asymmetry: nasals are handled robustly and stop loosely, the same split Table 5 found surviving the period gap, appearing here in what

the scribe chose to write rather than in what survived. Two independent measurements of one underlying fact.

8.3 When a consonant is written twice, was it said twice?

A scribe could spell one source consonant across a syllable boundary: the first character ends in `-t`, the second begins with `t-`. Whether doubling records a geminate or is a convention of the script changes how many consonants a transcription claims, and therefore how a name may be read. It is answerable on the Sanskrit corpus, where 772 pairs align one character to one syllable.

We take every junction at which a written stop or nasal coda is followed by a same-place onset, and ask whether the Sanskrit form carries that consonant once or twice:

JUNCTIONS WHERE A WRITTEN CODA IS FOLLOWED BY AN ONSET OF THE SAME PLACE	THE SOURCE WORD HAS THAT CONSONANT ONLY ONCE	THE SOURCE WORD HAS IT TWICE
140	**121 (86%)**	19 (14%)

TABLE 8 The double spelling is a convention, not a geminate: 釋迦 for *śākya*, 怛他 for *tathā*, 富單 for *pūtana*. A sequence such as 骨都 `kuət tɑ` therefore transcribes a source *kuta* with one `t`, not *kutta*. The count is conservative: "twice" is scored whenever the doubled letter appears anywhere in the Sanskrit form, so a geminate elsewhere in the word counts against the merger, and 86% is a floor rather than a ceiling. Like §8.2, this is a measurement that settles an objection which would otherwise be argued from intuition: a reader who assumes two written consonants mean two spoken ones will reject readings that the corpus permits. The nasal case is cleaner still: every source word in the corpus with a medial `ŋg` or `ŋk` is written with a `-ŋ` coda followed by a velar onset, 13 of 13: 央具梨 for *aṅguli*, 商羯羅 for *śaṅkara*, 僧伽施 for *sāṅkāśya*. Two characters spelling one medial nasal is not a license the scribe occasionally took; it is the only device available to him, since a Chinese syllable cannot begin with `ŋ`.

8.4 Which source vowel does a written vowel claim?

The three measurements above concern consonants. A reading proposal also stands or falls on the vowel, and where a character carries more than one Later Han reading it is often the vowel that separates the readings. 冒, the first character of the name §9.2 turns on, is such a character: `mək` with a central vowel in one reading, `mouᶜ` with a rounded one in the other. So we put the same question to the vowels that §8.2 put to the codas, on the same 770 aligned pairs. Each character is paired with the source vowel in its own position, and the source vowels are grouped into the three classes Later Han shares with Sanskrit, plus the vocalic `ṛ`, which it does not have.

SOURCE VOWEL	N	WRITTEN OPEN A ɑ	WRITTEN CENTRAL ə ɪ	WRITTEN FRONT I E	WRITTEN ROUNDED U O ɔ
open *a ā*	1509	**1194 (79%)**	105 (7%)	142 (9%)	65 (4%)
front *i ī e ai*	387	23 (6%)	19 (5%)	**322 (83%)**	22 (6%)
rounded *u ū o au*	294	80 (27%)	10 (3%)	31 (11%)	**173 (59%)**
vocalic ṛ	15	3 (20%)	4 (27%)	5 (33%)	3 (20%)

TABLE 9 The diagonal holds for the three classes the two languages share. The interest is off the diagonal, and it is asymmetric. A written rounded vowel is the reflex of a source rounded vowel in 59% of cases and of a source open vowel in 4%, about fourteen to one, so a rounded spelling carries real information about the source. A written central vowel is the reflex of a source open vowel in 7% of cases and of a source rounded vowel in 3%, so a central spelling carries very little, and the gap between those two figures is not one a reading should be asked to rest on. The last row is the control for a source vowel the writing system has no counterpart for: it scatters across all four classes with no majority, on fifteen cases. Neither language in this corpus has a front rounded vowel, so the substitution §9.2 needs cannot be priced in period. Out of period it can: in Ligeti's Ming glossary a source *ö* or *ü* is written with a rounded Chinese syllable 58 of 60 times, 97%, and with a central one once. We use that as a cross-period bound and not as a Han measurement.

8.5 Did the scribes hear the difference between Turkic *q* and *k*?

Turkic has two velar stops in complementary distribution with the vowel: a back `q` before *a, ı, o, u* and a front `k` before *e, i, ö, ü*. Chinese had no such rule, but it did have a plain and an aspirated velar, and a fricative `h-`. Whether scribes used those to render the Turkic distinction has not, so far as we have found, been counted. It cannot be counted on the Later Han corpus, because Sanskrit has neither the distribution nor a uvular at all. It can be counted on Ligeti's Ming glossary, where the source language is Turkic and the distinction is present in every word.

TURKIC WORD-INITIAL VELAR	ASPIRATED Kʰ-	H-	PLAIN K- OR G-	OTHER
back q (n=66)	20 (30%)	**43 (65%)**	2 (3%)	1
front k (n=35)	**31 (89%)**	2 (6%)	1 (3%)	1

TABLE 10 The two are written differently, and the majorities point in opposite directions: a back q takes an h- character about two times in three, a front k takes an aspirated one about nine times in ten. The scribes were not treating the Turkic velar as one sound. **The caution §7 makes in general terms applies to this table in particular.** It is measured on a Ming glossary, not on Han material, and we use it as a cross-period bound in the same way, and for the same reason, as the front rounded vowel in Table 9: the Han corpus contains no instance of the sound to be priced. What it licenses is a statement about Turkic transcription practice, not about the second century BCE. Its practical use is that it separates two readings of a polyphonic character: where the table gives a character both an aspirated and an h- value, the choice between them is no longer arbitrary.

9 A worked example: what a reading proposal must survive

Section 6.3 shows that reading a transcription against a dictionary is, on its own, at chance. That is not an argument that no such reading can ever be supported; it is an argument that support must come from somewhere other than the resemblance. One case in this material meets that standard, and it is set out here as a worked example rather than as a result about Xiongnu.

From 呼韓邪 onward, every chanyu title ends in 若鞮, Later Han ńak te. *Hou Hanshu* states that the element was added to the title from that reign and glosses it 孝 "filial". This is the only Xiongnu title element the Chinese sources translate, which means the meaning is *recorded*, not inferred from the sound.

Four things then must hold, and each is checkable independently of the others.

- **The sense must be attested.** We searched the two lexicons of §3.4 and found *Inak*, glossed *fidelis* "faithful", in the *Codex Cumanicus*.[14] It is absent from Kāşgarî, so the word belongs to the Kipchak tradition. Old Turkic *ınaq* "trusted one, confidant" is generally derived, with *ınan-* "to trust", from a root *ına-* "to rely on". The sense the proposal needs is documented in a medieval glossary, and it matches 孝 independently.
- **The word is of the kind that becomes a title.** This matters more than the gloss alone. Clauson records Chagatay *inak* as *nāyib va muqarrab*, "a royal representative or senior minister", and the cognate *ınanç*, from the same *ına-* root, appears in Old Turkic as an office and as a name element: *él inanç tirek* in a Uyghur list of proper names, *ınançları buyrukları* "his ministers and officers", *inanç tayanç* for a confidential minister to a *beg*, and *él ögesi ınancu bilge* in the Yenisei epitaphs of southern Siberia.[4] A court title built on this root is therefore not a stretch imposed on the word; it is what the word ordinarily does. *Hou Hanshu* has 若鞮 entering the formal name from a particular reign onward, which is exactly how such a title spreads.
- **The morphology must be licensed.** A reading *ınaq-tı* is not available: *-tI* is not a noun-forming suffix, and the lexicons show it rather than a grammar having to assert it. Of the 31 Cuman headwords ending in *-tI*, almost all are glossed with Latin perfects (*Ayti: dixit*, *Azti: transgressus est*, *Öwtiti: detersit*), the definite past ending; the remainder are numerals or Arabic loans. Kāşgarî's index has six, none containing a suffix. A title cannot be a past tense verb. Old Turkic does have a suffix that fits: *-t*, a plural used specifically with *titles*: *tegin* → *tegit*, *tarqan* → *tarqat*.[7] On that reading the element is **İnakt**, "the faithful ones", and 若鞮 is not one ruler's epithet but a title borne by a line, which is exactly how the Chinese sources use it.
- **Objections to the spelling must be tested, not asserted.** Against a final -t it can be objected that Later Han had -t final syllables and a scribe would have used one rather than the open 鞮 te. That is a claim about scribal habit, and

Table 7 counts it: 30% of stop-final source syllables were written with an open character. Hence the objection does not hold.

- **What remains unexplained must be stated.** If the source was *ınakt*, the opening *ı-* is not written, and **若鞮** has no room for it. That objection stands.

Three of four hold, with the fourth named. That is the standard this paper asks of a reading proposal, and it is deliberately higher than resemblance. The sense is attested in a source contemporary with neither party to the comparison, the morphology is licensed by a suffix with the right distribution, and the one objection that could be tested was tested and failed.

One further name, **冒頓**, is treated separately in §9.2, because there the frame does not select a single reading and what the measurements yield is the price of each branch rather than a proposal. We produced 31 others in the course of this work. None of them meet it (by §6.3 they are at the chance rate), and they are therefore **not presented here as findings**. We include them as **Supplementary Material S1** and release them in the repository as `author_proposals.csv`, dated and labeled, because readers will run this search for themselves regardless and it is better that the output is written down than imagined.[26]

We supply S1 for the record, not for review. It has the status of a laboratory notebook that happens to be legible. Each row gives the Chinese form, the Later Han reading, the author's rendering, the sense proposed for it, and, where one exists, the historical or onomastic parallel that suggested it. Those parallels are drawn mostly from later Turkic naming practice, where a ruler's epithet is a common noun (Bayezid *Yıldırım* "the thunderbolt", Alp Arslan "the lion", Mehmed *Avcı* "the hunter"), and they are offered as the kind of thing that would have to be argued rather than as an argument. A reader who accepts none of them loses nothing from the paper. The measurements in §§5–8 do not rest on any of them. The one reading that is argued here, *İnakt*, is in S1 too, marked as the exception, so that the file can be read as a single list with the standard visible in it. S1 also contains a worked demonstration, at §S1.5, of why the decoder's own output must not be quoted in support of an individual reading, the concrete form of the warning in §6.2.

9.1 Two cases which are not ours to propose

We can apply the same standard to comparisons somebody else has already made. The selection bias §6.3 measures is removed by construction if the reading was proposed by others. Two items in the record carry a reading from the literature, one of them with a Chinese gloss to check it against.

The first is 徑路 `keŋᶜ laᶜ`, the **徑路刀**, a Xiongnu blade used in oath-taking. Friedrich Hirth compared it with Turkic *kıŋrak* in 1908, working from the *Hanshu* account of the treaty of 47 BCE, where the scholiast glosses the word as the precious sword of the Xiongnu. He drew the Turkic side from living Turki dialects rather than from Kāşgarî, and extended the identification to **輕呂**, the word the *Zhoushu* uses for King Wu's dagger, which is why he called it the oldest Turkic word on record. That further claim is his and is not tested here.[11] Kāşgarî glosses *kıñrak* as a large knife resembling a cleaver, for cutting meat and dough.[12] The Chinese side independently says the referent is a blade, two sources with no contact agreeing about what the object is, which is a stronger kind of agreement than a sound match. The consonant frame is exact: **徑** gives the velar onset with its `-ŋ` coda written, and **路** gives a Chinese `l-` for a source `r-`, the direction attested in 12 of 14 cases. The Turkic word's final `-k` is not written, and Table 7 gives the rate at which a stop-final source syllable was spelled with an open character: 30%.

Where the Turkic word comes from. Clauson derives *kıŋra:k* as a deverbal noun from an unattested **kıŋra:-* (the colon is Clauson's mark for a long vowel), itself denominal from *kıŋır*, in the sense of something curved or something that cuts crookedly.[4] Kāşgarî's own index supports that family on the same frame: *kıñır*, *kıñru* "askew, squint", *kañrak* "palate", the arch of the mouth, and the verb *koñur-* "to wrench, bend back", surviving as Turkish *kanırmak*; and the *Codex Cumanicus* independently glosses *Kingir* as *curvus*. A curved blade named from bending is therefore motivated inside Turkic by the standard authority, not by us.

The word also survives. Clauson records it in Teleut as *kıŋırak*, and in Kirghiz as *kıŋarak / kıŋırak*, "a rough two-edged knife used for cutting felt, scraping hides and sheepskins", the same object, in living languages, a thousand years after Kāşgarî and two thousand after the *Hanshu*.

Kāşgarî's text at volume III, page 382 of the *Dīwān* reads *kırpa:k* and is corrected in the margin to *kıŋra:k*. So this form rests on a marginal emendation rather than the text as written. The emendation is not the only support for it: Hirth took the word from living Turki dialects in 1908, and the Teleut and Kirghiz survivals cited above are independent of the Dīwān's manuscript altogether. *kıñır*, on the other hand, fits the transcription exactly, at three consonants with nothing supplied; *kıñrak*, which means a knife, needs the unwritten final `-k` that Table 7 licenses at 30%. The form that fits best does not mean "blade", and the form that means "blade" does not fit best.

> **A rival account.** The same object has been derived instead from an Iranian word for a short sword, the type whose Greek name is *akinakes*. The history supports it: the Xiongnu's western neighbors were Iranian-speaking, and the weapon is present in steppe archaeology, so a borrowed name is not a stretch.
>
> This is testable with the same machinery, and does not survive the test as well. 路 `lɑᶜ` is a liquid-onset character, so the channel requires a liquid in the second position of the source word. The *akinakes* family (`k–n–k`) has no liquid anywhere in it. If the proposal is that 徑路 transcribes that word, it fails on the frame, not at the margins.
>
> **We could not trace the Iranian derivation to a source.** It circulates in the secondary literature on this word, but the works we checked give no etymon, and Hirth, who is cited nearby for the Turkic comparison, does not mention the Iranian account at all. The test above therefore applies to the *akinakes* family, which is the form the account is usually stated in, and not to a specific etymon that someone has actually put forward. If a reader can supply the reference the test can be rerun against it in an afternoon, since the machinery is in the repository. In any case our channel establishes what a transcription *permits*, never what it requires: a Turkic reading surviving does not make a Turkic source the only one available. This is Doerfer's objection in its sharpest form, and the third case below is what it looks like when the objection bites. The honest report is that the Turkic candidate survives its test and the Iranian one, as usually stated, does not.

The second is 閼氏 `ʔɑt dźeᴮ`, the title of the chanyu's principal consort, and it is the case where the channel returns a verdict that does not favour the language this paper spends most of its pages on. Vovin, arguing that the Xiongnu spoke a Yeniseian language, rejects Pulleyblank's identification of the title with Turkic and Mongolian *qatun* on two grounds: that it rests on a single textual example in which the first character is read with an initial `y-` rather than a glottal stop, and that neither Turkic nor Mongolic `q-` was a voiced fricative at that date.[19,28] In its place he offers Proto-Yeniseian `*ʔal(V)it-` "woman, wife", surviving as Kott *alit*, Assan *alit* and Arin *biqam-alte*, and a Xiongnu *alte* or *elte* which, he writes, could be transcribed in Chinese only as `*ʔat-tej` or `*ʔen-tej`, since the Old Chinese syllable-final `*-r` was no longer extant in Former Han times.

The transcription argument is correct, and it is the one §8.2 measures. Later Han had no liquid coda, so a source syllable ending in one was written with a `-t` or left open: of 37 such syllables in the Sanskrit corpus, 21 were left open and 14 took a `-t`, 38%. Vovin's reading is well formed on that count. It uses the written `-t` of 閼 rather than discarding it, and the corpus licenses the step at an ordinary rate. His alternative `*ʔen-` is weaker: both reconstructions we can check give 閼 a single reading with a stop coda, `ʔɑt` in Schuessler's table and Middle Chinese `ʔat` from Old Chinese `*qˤat` in Baxter and Sagart.[2,22] The `-n` belongs to 焉, the different character used in the parallel spelling 焉支, so the `*ʔen-` reading is an inference from an alternation in usage rather than a value any reconstruction assigns to this character.

What does not follow is the language. The segments Vovin reconstructs are also a Turkic word. Modern Turkish *elti* is what each of two brothers' wives calls the other, and in dialect what two wives of one man call each other, and it has the same shape as his *elte*: the same liquid that has to be written with the `-t`, the same dental in the second syllable, the same final vowel. A scribe writing *elti* would have produced 閼氏 by exactly the steps Vovin sets out for *elte*. His "only" is a claim about Chinese orthography and it is true; it is not a claim about which language the source was, and the spelling does not make one.

Nor are the two words as differently placed in time as their presentation suggests. Clauson records *élti:* glossed "lady", standing opposite *al-jāriya* "slave girl", in a fourteenth-century Arabic-Turkish glossary, so the Turkic word is attested medievally in the sense a consort's title would want and not only in its modern kinship use.[4] We did not find it in the Türk Dil Kurumu index to Kāşgarî or in the *Codex Cumanicus*.[12,14] Vovin's side rests on Kott, Assan and Arin forms recorded in the eighteenth century and later, from languages with no early texts, carried back by reconstruction; he notes himself that the proto-meaning "woman" is reconstructed for the sake of a North Caucasian comparison, and that "woman" occurs in Kott alone while "wife" occurs in two languages. Neither word reaches the second century BCE by attestation. What the channel can say is what the spelling permits, and it permits both.

So, three comparisons from the literature have been tested rather than generated. Two survive tests that 32 of our own readings fail; one of those two has a rival etymology our method cannot rule out; and the third is well formed but does not identify a language, which is what §6.1 predicts. That spread is the point of this section. The channel cannot generate supported readings at this data density, but it can judge ones already on the table, including ones advanced against the hypothesis this paper spends most of its pages on, and deciding on other people's proposals is a use of it that the accuracy figures actually license.

9.2 Where a century-old disagreement actually lies: 冒頓

冒頓 is the most consequential name in the record, and the frame it fixes looks unusually tight, because **both codas are written**. What makes it hard is that neither character has a single value. Schuessler's table gives 冒 two Later Han readings, `mək` and `mouᶜ`, and 頓 two, `tuənᶜ` and `tuət`, one ending in a nasal and one in a stop.[22] A third value for the second character also circulates. A commentator notes 頓音毒, "頓 is pronounced 毒", which Pulleyblank reports as Early Middle Chinese `dawk`.[20] That needs stating precisely, because published readings of this name rest on it. In the Later Han table 毒 and 纛 are both `douk` while 頓 is `tuənᶜ` or `tuət`. A velar stop coda is therefore not a row of 頓 at all; it is a substitution recorded by a commentator. We keep it in view as a third value on offer, labeled for what it is, because the alternative is to let a reading depend on it silently.

The first character is not where the difficulty is. In the Later Han table 冒 and 默 share the reading `mək`. And 默啜, the personal name of the Türk ruler known in the inscriptions as Qapaghan Qaghan, is read *Bögü-Çor*. So that exact Later Han syllable is on record writing *bögü* in a Turkic ruler's name: the `m-` standing for the source `b-`, the written `-k` standing for the source `g`, and the final vowel of *bögü* not written at all. The substitution is on record more than once in Turkic onomastics, and the characters used are worth setting out: 默 `mək` in 默啜 for *Bögü-Çor*, and 牟 `mu` in 牟羽 for the Uyghur Bögü Qaghan. Both write *bögü* with a Chinese `m-` character, and the two fall either side of 冒's own split: 默 matches its first reading exactly, and 牟 is the nearer of the two to its second. Whichever reading of 冒 is taken, there is a Turkic precedent for that character writing *bögü*. Both authorities in fact give this name the unrounded reading: Pulleyblank lists the character at K1062a and glosses the entry "covet; Modun, founder of the Xiongnu empire".[20]

Table 9 prices the two readings against each other. A rounded source vowel is written with a rounded Chinese vowel in 59% of cases and an open source vowel in 4%, so the rounded reading of 冒, `mouᶜ`, is where a rounded first element is ordinary and an open one is not, by about fourteen to one. The unrounded reading, `mək`, separates the two candidates far less: 7% against 3%, a gap too small to carry a reading, and the one attested case of this syllable writing *bögü* sits on that very reading. Each step of the first character is separately licensed. The `m-` for a source `b-` is 3 of 22 in the Later Han layer, 13.6% (§8.1). One character covering two source syllables, with a written stop coda doing the compressing, occurs in 47 of the 1,017 verified pairs: 竭 for *-gata* in *tathāgata*, 薩 for *-sattva* in *mahāsattva*, 弗 for *-putra* in *śāriputra*. 冒 for *bögü* is that same shape.

The second character decides the name, and each of its three values admits a different shape of second element. The corpora say how often a source coda of each class is written with a coda of the class each value carries:

VALUE TAKEN FOR 頓	WHAT IT IS, AND WHO ASSIGNS IT	A NASAL-FINAL SECOND ELEMENT	A VELAR-FINAL SECOND ELEMENT, *TUĞ*	*BAΓATUR*
`tuənᶜ` dental nasal coda	The first row of Schuessler's table, and the character's primary reading. It is the value Appendix A prints, here as everywhere. Schuessler's own note for this name points to his second row instead, so taking the primary reading here is a position rather than a default.	**fits** **77 of 93 (83%)** the commonest cell in the corpus	**fails** 0 of 21 the one nasal spelling of a source velar is a velar nasal, not a dental one	**fails** 1 of 37 (3%) a source `-r` against a written `-n`
`tuət` dental stop coda	The second row of the same table, and the reading Schuessler assigns to this name, at pp. 209 and 361.	**fails** 0 of 93 a source nasal is not written with a stop coda	**permitted, uncommon** 2 of 21 (10%) the place changes, the manner does not	**fits** **14 of 37 (38%)** the scribe's usual choice for a source `-r` after an open spelling
the substitution `douk`, velar stop coda	Not a row of 頓 at all. It is the syllable 毒 substituted by a commentator, 頓音毒, reported by Pulleyblank as Early Middle Chinese `dawk`. A tradition about how to read the character, which is a different kind of claim from a reconstruction of it.	**fails** 0 of 93	**fits** **10 of 21 (48%)** both codas written, same place, nothing discarded	**fails** 0 of 37 a source `-r` is never written with a velar coda

TABLE 11 Each value of 頓 admits a different second element, and none admits two. Counting the values that license each candidate: **two of the three admit a velar-final element, one admits *Bayatur*, and one admits a nasal-final element**. The rates are the rate at which the Later Han corpus writes a source coda of that class with a coda of the class the value in that row carries; the nasal and velar columns come from the same tally as Table 7, the liquid column from §8.1. Denominators differ because the corpus offers different numbers of opportunities, and the velar column rests on 21 cases, so it should be read as an indication rather than a rate. *Bayatur* has been the field's reading for a century and is licensed on the dental stop reading alone, which is also the reading Schuessler names for this name. A velar-final reading is licensed on the other two. That is the disagreement, and it is a disagreement about which value to take, not about phonology.

> **The disagreement is smaller than it looks, and it is not phonological.** Readings of the founder's name have stood against each other for a century. They do not rest on different views of Later Han phonology, of what a Chinese scribe would do, or of Turkic word formation. They rest on which value is taken for one character, and the transcription evidence cannot choose, because the values describe the same character on the same page, or in the case of the third, in the same commentary. What the channel contributes here is not a reading but the location of the disagreement and the price of each branch.
>
> The conventional pronunciation of the compound, *Mòdú*, points to a stop rather than a nasal. On the table above that excludes the nasal-final shape and admits both of the others, which is as far as it goes. It is a reading tradition rather than a measurement, and it is not ours to adjudicate.

What each branch has to assume. On the standard of §9 all three shapes clear the morphological test, since two nouns in apposition is the commonest Turkic name shape, as in Alp Arslan or *Bögü-Çor* itself. They differ in what else has to be granted. The liquid-final shape needs the dental stop reading `tuət`, and it needs a word: Clauson's entry on *bayatur* records that the word "occurs only once in the early period and then still as a P.N.", that the sense "picked warrior" most probably developed in Mongolian rather than Turkic, and that Turkic had other words such as *alpayut* in that meaning.[4] The velar-final shape needs either `tuət` or the substitution, and its candidate second element is a well attested Old Turkic noun, but the direction of its loan relation with Chinese is disputed, which we set out in S1 rather than here. The nasal-final shape needs the nasal reading `tuənᶜ`, where it is the ordinary spelling, and it needs a vowel the frame does not recover, since the corpus fixes neither the quality nor the length of the vowel in the second syllable.

The two steps priced together. Table 11 prices only the second half of the name. Each branch also needs a first step, and it is the same step on all three: on any Turkic reading 冒 `mək` writes a source *b*-, which the Later Han corpus does in 3 of 22 cases, 13.6% (§8.1). Carrying that through the coda rates gives the cost of the whole name rather than of its second element. A nasal-final second element on `tuənᶜ` comes to 11.3%, about 1 in 9. A velar-final element on the 毒 substitution comes to 6.5%, about 1 in 15. *Bayatur* on `tuət` comes to 5.1%, about 1 in 20, and *Bayatur* on `tuənᶜ` to 0.4%, about 1 in 270. These joint figures cannot be written as a count, because the two rates are drawn from different tallies with different denominators, and they assume the two steps are independent, which is an assumption and not a measurement. Note also what the first step does not do: because the *m*- rate is common to every Turkic branch it scales all of them alike, so it changes how a Turkic reading of this name stands against a non-Turkic one, and does not reorder the rows of Table 11.

We do not choose among them, and the reason is not modesty. The three branches are not three hypotheses about the same evidence; they are three different pieces of evidence, and the choice between them is a decision about Chinese philology, not about transcription practice. What this section offers is the priced comparison. The author's own proposal for this name, with the objections to it recorded alongside, is in **Supplementary Material S1**, which is where proposals in this work are kept.

The choice is not peculiar to this name. Across the record, **46 of the 107 character slots the table covers carry more than one Later Han reading, and in all 46 the value printed in Appendix A is the first tabulated one** (§11). 冒頓 is the case where that choice carries the most weight, and it is the case where the first tabulated reading is not obviously the right one. The material for choosing, the counts, the corpora and the scripts, is in the repository that we provided.

10 Where the method does work

The constraint identified in §5 is data density, not the difficulty of the language. That suggests where the future work needs to go next. Göktürk-era material has what the Xiongnu material lacks: a source language that is not in dispute, an independent script recording it, and enough of it to train on.

CHINESE, AS WRITTEN IN THE TANG HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	INDEPENDENTLY IDENTIFIED AS, IN ESTABLISHED SCHOLARSHIP
葉護	jap ɣuaᶜ	yap ğua	*yabghu*, a princely title
可汗	kʰaiᴮ ganᶜ	kai gan	*qayan*
特勤	dək gɨən	dık gıın	*tigin*, prince
突厥	tʰuət kyat	tuıt kyat	*Türk*
阿史那	ʔa ʂiəᴮ na	’a şiı na	*Ashina*, the ruling clan
黠戛斯	gɛt kɛt sie	get ket sie	*Qïrqïz* (Kyrgyz)

TABLE 12 **Neither middle column is model output.** Column two is Schuessler's published Later Han reconstruction applied to Tang-era words, deliberately the wrong

period, which is the point of the exercise. The last column is not an output of this work either: each identification is long settled on evidence independent of the Chinese transcriptions, the Orkhon inscriptions giving *qayan*, *tigin*, *yabghu* and *Türk* directly in Old Turkic script, several corroborated again in Sogdian, Arabic and Byzantine sources.[7,9] They function here as ground truth. That 葉護 → *yap ǧua* is readable against *yabghu* despite a period mismatch of four centuries measures the conservatism of Chinese scribal practice, not linguistic continuity between the Xiongnu and the Göktürks, on which this material bears not at all.

Chinese records name a great many steppe peoples, and they fall into three groups for these purposes. **Tractable**: those recorded in the Tang histories whose language is known: 鐵勒 Tiele, 回紇 Uyghur, 葛邏祿 Qarluq, 突騎施 Türgesh, 黠戛斯 Kyrgyz, alongside the Göktürk material. These have transcriptions, a securely identified source language and published readings for the right layer; they are training data. **Readable but uninformative**: groups such as 丁零 Dingling and the Avars, whose Chinese readings can be recovered but whose language is unknown, so a reading cannot be checked. **Out of reach**: the Xiongnu themselves, and the Pechenegs, who are barely in the Chinese record at all. The distinction is not how interesting the group is but whether there is anything to validate against.

11 Limitations of this work

- **No independent philological review.** The computational method and its evaluation are ours to defend. Philology is not our work. The Later Han readings are Schuessler's, the etymologies discussed are from the published literature, and the identifications in Table 9 are the established consensus.
- **Corpus validation is internal.** The extraction rules are documented and internally consistent (93.4% length agreement for Mongolian, an induced akṣara inventory matching the standard one for Sanskrit), but no external reviewer has checked a sample against the sources.
- **The Turkic corpus is Ming-era**, roughly 1,200 years later than the Xiongnu material. §7 measures what those costs rather than assuming it away.
- **Reconstructions are inferences, and the period gap runs on the Chinese side too.** Schuessler, Pulleyblank and Níng are three scholars with three notations, and Baxter and Sagart a fourth.[2,19,20] It is tempting to assume that the disagreement between them is notational, and we measured whether it is. The names are recorded in the *Shiji* and the *Hanshu* for events of the third to the first century BCE; the table this paper reads them with describes Chinese of the first and second centuries CE. Comparing Schuessler's Later Han against Baxter and Sagart's Old Chinese over the record's 81 distinct characters, 60 of which appear in both, **38 carry the same class of segment in both and 22 do not: 16 differ in the onset, 4 in the coda, and 2 in both, 37% in all**. That is not a notational difference. It is a difference in what the character could write. 于 is a glide in the later table and a velar in the earlier one; 單 ends in a nasal in the later and a liquid in the earlier; 路 is open in the later and closed with a velar stop in the earlier. The two reconstructions bracket the transcriptions rather than dating them, so this is the width of the uncertainty and not a correction to Appendix A, but it is a real cost and every reading in this paper carries it. Table S8 of the Supplementary Material lists the 22 characters. A single table is also not single-valued: **of the 107 character slots the table covers, 46 carry more than one Later Han reading**, and Appendix A prints the first tabulated row in all 46. The record has 114 slots in all, 81 of them distinct characters, and 7 slots are not in the table at any reading. Where the alternative would change a reading it is named in the entry, and §9.2 is the case where it changes most.
- **Names travel.** Even a perfect reconstruction would not settle ethnicity.[16,21] Turkic tribes bore Iranian names and Mongol rulers used Turkic titles; the caution in §6.3 about *tengri* is a specific case of this general problem.
- **Reproducibility has a software dependency.** The headword count for the Kāşgarî lexicon differs between machines, 6,820 against 6,880, according to

the version of the PDF text-extraction library. It shifts the noise floor by 0.008 and changes no rank or conclusion.

12 Data and code availability

We package everything described here as one repository, so that anyone can regenerate every number and table in this paper from the raw inputs by running the numbered scripts in order. The repository is the durable output of this work and this paper describes it.[26]

```
chinese-transcription-channel/
├── README.md            what each output is, and how to regenerate it
├── LICENSE              code MIT
├── DATA_LICENSE.md      the three-way split described below
├── data/derived/        the corpora, the lexicons, and every measurement
├── src/                 steps 1–40 plus lookup_lexicons.py, each rerunnable alone
├── reports/             one summary per step, with the numbers quoted above
└── docs/                design notes, including approaches that were rejected
```

Supplementary Material S1. *Candidate readings of the Xiongnu record* accompanies this submission and carries the same content as `data/derived/author_proposals.csv` in the repository. We supply it for the record under §9 and do not offer it as a result of this paper.

Released, CC BY 4.0. The Mongolian and Sanskrit corpora, the Cuman lexicon (Kuun's 1880 edition being public domain), the induced akṣara table, the Xiongnu record of Appendix A, and every measurement file behind Tables 1–12, including `period_depth.csv`.

Withheld, with the extraction script released instead. Five derived files inherit restrictions from their sources: the Turkic pairs, from Ligeti's glossaries (© Akadémiai Kiadó); the Kāşgarî lexicon, from a modern Türk Dil Kurumu edition; two files built on that lexicon, the Old Turkic echo results and the candidate-space listing, the latter also quoting Clauson's *Etymological Dictionary* verbatim in a context column; and a word list of the *Irk Bitig*, from Tekin's 1993 edition (© Harrassowitz), which we use only to date a proposed reading in the Supplementary Material and which no measurement in this paper depends on. Anyone holding the sources can regenerate any of them byte-for-byte by running the numbered script, which keeps the paper's numbers checkable without redistributing the source. No permission is required for this arrangement, and none has been sought.

Not in the repository. Third-party source texts and datasets (dictionaries, scanned volumes, Unihan, the DILA authority files) are inputs, not outputs. Where a script needs one, its docstring says which file and where to obtain it.

A Appendix: The Xiongnu record

All thirty-nine names, titles, and lexical items in the Chinese histories, with the Later Han reading of each character and a Turkish-orthography rendering for readers who do not read phonetic notation. The Later Han column is published scholarship, not model output; our contribution is the assembly: we collected the names in one place and joined the readings to them character by character. Superscript letters mark Han tone categories. A dagger, †, marks a **polyphonic character**, one for which Schuessler's table gives more than one Later Han reading; the first tabulated row is the one printed. **46 of the 107 character slots the table covers are so marked**, and where the alternative would change a proposed reading it is named in the Supplementary Material rather than left to the dagger. Table S7 of the Supplementary Material lists every polyphonic character together with the readings not used. 若鞮 is listed once, shaded, rather than repeated in the six titles that carry it.

A.1 Rulers and the ruling clan

NAME OR CLAN NAME, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	DATES OF RULE, AND THE SOURCE IN WHICH THE FORM APPEARS
頭曼 Touman	do manᶜ †	do man	209 BCE

NAME OR CLAN NAME, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	DATES OF RULE, AND THE SOURCE IN WHICH THE FORM APPEARS
冒頓 Modu / Modun	mək† tuən^{C}†	mık tuın	209–174 BCE
稽粥 Jizhou / Jiyu	kʰei^{B}† tśuk	kei çuk	174–160 BCE
軍臣 Junchen	kun dźin†	kun cin	160–126 BCE
伊稚斜 Yizhixie	ʔi ḍi^{C} ja	’i di ya	126–114 BCE
烏維 Wuwei	ʔɑ wi	’a vi	114–105 BCE
詹師廬 Zhanshilu	tśam ṣi liɑ	çam şi lia	105–102 BCE
呴犁湖 Xulihu / Goulihu	huo† lei† gɑ	huo lei ga	102–101 BCE
且鞮侯 Qiedihou	tsɑ^{B}† te† go	tsa te go	101–96 BCE
狐鹿姑 Hulugu	ɣuɑ lok kɑ	ğua lok ka	96–85 BCE
壺衍鞮 Huyandi	gɑ jan^{B}† te†	ga yan te	85–68 BCE
虛閭權渠 Xulüquanqu	kʰɨɑ liɑ gyan gɨɑ	kıa lia gyan gıa	68–60 BCE
握衍朐鞮 Woyanqudi	ʔɔk jan^{B}† guo te†	’ok yan guo te	60–58 BCE
呼韓邪 Huhanye	hɑ† gɑn ja†	ha gan ya	58–31 BCE
郅支 Zhizhi	tśit tśe†	çit çe	56–36 BCE
若鞮 Ruodi	ńɑk te†	nyak te	title element
復株累若鞮 Fuzhulei Ruodi	buk† ṭio lui† ńɑk te†	buk tio lui nyak te	31–20 BCE
搜諧若鞮 Souxie Ruodi	so† gɛi ńɑk te†	so gei nyak te	20–12 BCE
車牙若鞮 Cheya Ruodi	tśʰa† ŋa ńɑk te†	ça nga nyak te	12–8 BCE
烏珠留若鞮 Wuzhuliu Ruodi	ʔɑ tśo liu ńɑk te†	’a ço liu nyak te	8 BCE–13 CE
烏累若鞮 Wulei Ruodi	ʔɑ lui† ńɑk te†	’a lui nyak te	13 CE–18 CE

NAME OR CLAN NAME, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	DATES OF RULE, AND THE SOURCE IN WHICH THE FORM APPEARS
呼都而尸道皋若鞮 Huduershi Daogao Ruodi	hɑ† tɑ ńə† śi douᴮ† kou ńɑk te†	ha ta nyı şi dou kou nyak te	18 CE–46 CE
攣鞮 Luandi	lyan te†	lyan te	–
虛連題 Xulianti	kʰɨɑ lianᴮ† de†	kıa lian de	–

Twenty-one *chanyu* from Touman to Punu, plus both forms of the ruling clan name: 攣鞮 in the *Shiji* and *Hanshu*, 虛連題 in the *Hou Hanshu*.

A.2 Titles and lexical items

TERM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	WHAT KIND OF ITEM IT IS: TITLE, LEXICAL WORD, OR ETHNONYM
單于 chanyu	dźan† wɑ	can va	title
撐犁孤塗單于 chengli gutu chanyu	ḍaŋ lei† kuɑ dɑ dźan† wɑ	dang lei kua da can va	title
撐犁 chengli	ḍaŋ lei†	dang lei	lexical
孤塗 gutu	kuɑ dɑ	kua da	lexical
閼氏 yanzhi / ezhi	ʔɑt dźeᴮ†	ʼat ce	title
屠耆 tuqi	dɑ gɨ†	da gı	title
谷蠡 luli / guli	kok† leᴮ†	kok le	title
當戶 danghu	tɑŋ† gɑᴮ	tang ga	title
骨都侯 guduhou	kuət tɑ go	kuıt ta go	title
且渠 juqu	tsɑᴮ† gɨɑ	tsa gıa	title
徑路 jinglu	keŋᶜ lɑᶜ	keng la	lexical
服匿 funi	buk† ṇɨk	buk nık	lexical
甌脫 outuo	ʔo tʰuɑt	ʼo tuat	lexical
胡祿 hulu	gɔ† lok	go lok	lexical

TERM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	WHAT KIND OF ITEM IT IS: TITLE, LEXICAL WORD, OR ETHNONYM
匈奴 Xiongnu	huoŋ nɑ	huong na	ethnonym
羯 Jie	kɨat	kıat	ethnonym

The lexical items are the only Xiongnu words the Chinese sources gloss. 撐犁 is glossed "heaven" and 孤塗 "son", which is why that pair has carried so much of the argument. The proposals for 孤塗 do not settle it. Pulleyblank compared it with Arin *bĭkjàl* "son"; Vovin holds that form to be isolated within Yeniseian, the rest of the family reflecting Proto-Yeniseian **puʔ-b* "son", and finds no Altaic alternative either.[27,28] The Turkic proposal, *qut* "fortune", fits the sound and not the gloss. So the one item where a Chinese gloss and a Yeniseian etymology were thought to meet is contested by the leading advocate of that hypothesis.

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